

12 (long)

(a) A capacitor consists of two concentric spherical metal shells of inner and outer radii r_1 and r_2 respectively with an air gap between.

(i) Using Gauss's law derive an expression for the capacitance. [9]

(ii) The electric field at which air breaks down is 3 MV m^{-1} . Assuming that $r_1 = 1 \text{ cm}$ and $r_2 = 3 \text{ cm}$ calculate the capacitor charge at which electrical breakdown of the air occurs. [3]

(b) A hollow toroidal ferrite ring with inner radius 4 cm and outer radius 8 cm with cross section shown in Fig. 7 is evenly wound with a 200 turn coil through which 1 A of current flows. The relative permeability of the toroid material μ_r is 500 and the hollow within the toroid is air-filled.

(i) What is the magnetic flux density B in the tangential direction through the ferrite in terms of r ? [4]

(ii) What is the magnetic flux density B in the tangential direction through the air core in terms of r ? [4]

(iii) What is the magnetic flux Φ in the air core and in the ferrite? [8]

(iv) What is the self-inductance of the coil? [2]

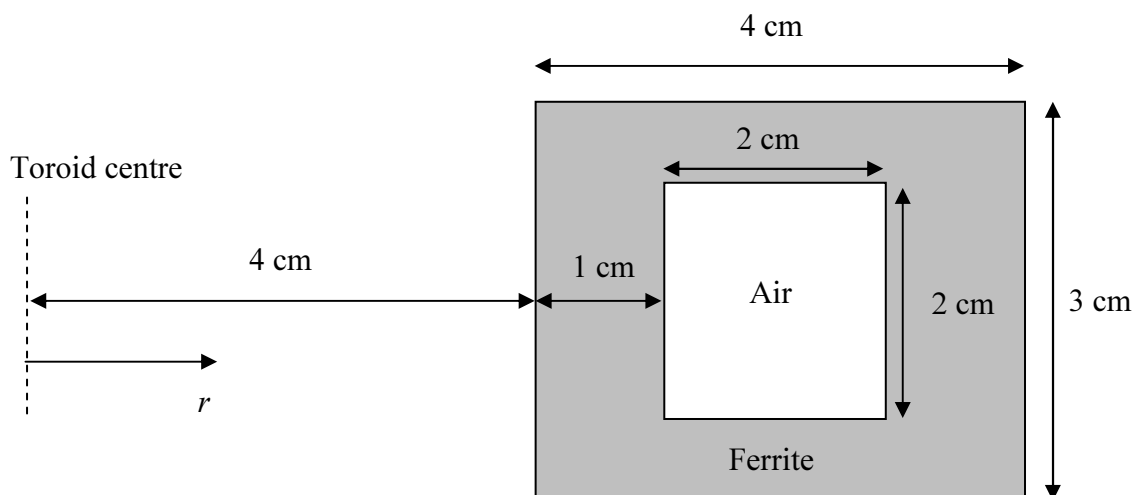


Fig. 7

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6 (b) $Z = \overline{B.C}$

7 (a) 8 kB, 4 GB (b) $\overline{CS} = \overline{A_{15}.A_{14}.A_{13}.Address Valid}$

8 (b) (i) C=1, DC=1, Z=1, 0x20 = 0F_h, W = 00_h (ii) C=0, DC=0, Z=0, 0x20 = 55_h, W = FF_h

9 (b) 2 bistables